



MicroWeigh Triple Indexer
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Contents

1	Introduction	5
1.1	MicroWeigh features	5
1.2	MicroWeigh applications	5
2	Keypad Operation	5
3	Main Menu Items	6
3.1	“Power off”	6
3.2	“Washdown”	6
3.3	“Totals”	6
3.4	“Calibrate”	6
3.5	“Setup menu”	6
3.6	“Audit cfg”	6
3.7	“Audit cal”	6
4	How to Step Through Menus	7
4.1	Menus can contain several different items	7
4.2	How to enter a number	7
4.3	How to select from a list	8
5	General Scale Operations	8
5.1	Scale on procedure	8
5.2	Scale off procedure	8
5.3	Zero procedure	9
5.4	Units procedure	9
6	Machine Operation	9
6.1	Initial setup	9
6.2	Starflex coordination	11
6.3	Startup	11
6.4	Shutdown	12
6.5	Cleanout	12
6.6	Target weight	12
6.7	Dump cycle	12
6.8	Dry contact output option	12
6.9	Claw option	13
6.10	Batching	13
6.11	Last box weight	13
6.12	Downstream jam	13
7	Calibration Procedure	13
7.1	Entering the calibration menu	13
7.2	Keying in cal weight	13
7.3	Calibration example	14
7.4	Establishing a zero	14

7.5	Accepting a cal weight	14
8	Scale Parameters	14
8.1	“Units”	14
8.2	“Capacity”	14
8.3	“Resltn”	15
8.4	“Stability”	15
8.5	“Motion sns”	15
8.6	“AZT”	15
8.7	“AZT band”	15
8.8	“Calibrate”	15
8.9	“IZ set”	15
8.10	“lb units”	15
8.11	“kg units”	16
8.12	“g units”	16
8.13	“oz units”	16
8.14	“Defaults”	16
9	Display Messages	16
9.1	General warning/error messages	16
9.2	Calibration warnings	17
10	Menus	18
10.1	Main level	18
10.2	Production	18
10.3	Totals	19
10.4	Setup menu	19
10.5	Parameters	20
10.6	Dump settings	21
10.7	I/O test	22
10.8	RS-485 menu	22
10.9	Info menu	23
11	Detailed Menu Item Descriptions	24
11.1	Main menu level	24
11.2	Production menu	24
11.3	Totals menu	25
11.4	Setup menu	25
11.5	Parameters	26
11.6	Info menu	27
12	Troubleshooting	28
12.1	Load cells	28
12.2	Banner opto sensors	28
12.3	Machines	28
12.4	Dump cycle state machine	29

12.5 Before calling WeighTech...	30
12.6 Module assignments (master indicator)	30
12.7 Module assignments (remote indicators)	30
13 Drive Settings	31
14 Drive Settings	35
15 Replacement Parts	39
16 Wiring Diagram	40
17 Load Cell Color Codes	45

1 Introduction

With the WeighTech MicroWeigh a combination of state-of-the-art technology with common down-to-earth basics creates a digital indicator that makes troubleshooting and actual maintenance repair so simple that anyone can be trained to make repairs on this indicator in just minutes.

1.1 MicroWeigh features

- High impact ABS alloy construction.
- Highly visible, easy-to-read display with adjustable contrast and backlight.
- Environmentally sealed touch-sensitive operator control panel.
- Standard units of measure include grams, kilograms, ounces, and pounds.
- RS-232 and Infrared communications are standard with RS-485 option available.
- Wireless data collection using a PDA with WeighTech ScaleTrax software.

1.2 MicroWeigh applications

- Standard weighing
- Label printing
- Tank or vat weighing
- Checkweighing (boxes, bags, and pieces)
- Bench and floor scales
- Batch weighing

2 Keypad Operation

The WeighTech MicroWeigh keypad is a watertight sealed touch sensitive sensor. The keys are actually sensitive to contact area, not force. Press lightly with the ball of your fingertip as though you were giving fingerprints. Best results come from using the ball of your finger, not the very tip. Most objects will not trigger the keypad—knives, screwdrivers, tools, etc. do not have enough surface area in contact with the key to register as a keypress. (You might get it to trigger with a medium sized conductive bolt head, if you have skin in contact with the bolt.)

One consequence of the design of the touch sensitive keypad is that it is sensitive to water streams. For this reason, WeighTech includes a unique “washdown mode” to prevent unwanted keypad activity during washdown/sanitation/cleanup intervals. When the indicator is in washdown mode, the indicator will weigh normally but the keypad is locked out.

To unlock the keypad, you must play follow the leader. One key will be lit. Press it. Another key will then light up. Press it. Continue until the indicator displays “Exit washdown”. The indicator will require that you press five keys in a row correctly before it will unlock the keypad. Any wrong keypress will restart the counter back to five. The odds are extremely slight that random water splashing would ever be able to trigger the correct keys in the correct order to unlock the keypad.

3 Main Menu Items

3.1 “Power off”

Touch the enter key to select this menu item, which will power down the indicator. If the auto-on jumper is installed on the interface board, the indicator will immediately turn back on.

3.2 “Washdown”

This function puts the indicator in washdown mode to prevent inadvertent keypad activity. See the washdown section of this manual for more information.

3.3 “Totals”

This function leads to the totals submenu.

3.4 “Calibrate”

This function allows you to calibrate the scale. Refer to the calibration section of this manual for details.

3.5 “Setup menu”

Enter the setup submenu, where scale parameters can be viewed or set.

3.6 “Audit cfg”

Displays the audit counter for configuration. Every time a sealed scale parameter is modified this counter will increment by one. This setting is nonvolatile (it will be retained even if the batteries go dead) and cannot be altered except by modifying an audited configuration parameter.

3.7 “Audit cal”

Displays the audit counter for calibration. Every time the scale is calibrated this counter will increment by one. This setting is nonvolatile (it will be retained even if the batteries go dead) and cannot be altered except by performing a calibration.

4 How to Step Through Menus

From the main weight display, press the “Menu/Help” key. You are now in a menu, and the keys now have different functions:

Cancel Help Enter Down Up

- “Cancel” will back you out of the menu one level at a time.
- “Help” will display information about the current choice (option).
- “Enter” has various functions, depending on where you are in the menu.
- The “Down” key will scroll backward through the menu choices.
- The “Up” key will scroll forward through the menu choices.

4.1 Menus can contain several different items

An item with a “*” on the right end will do something when you press the enter key—something might be turn the indicator off, drill down into another menu, clear totals, or start a calibration routine. The item with a numeric value (scale capacity, for instance) at the right side of the display might allow you to change the number by pressing the enter key. An item with text (such as “on” or “off”) at the right side of the display might allow you to select from a list of options by pressing the enter key. Some items are just for reference and cannot be changed at all. Examples of reference items would be the software name and revision—these are set when the software is written and cannot be changed.

4.2 How to enter a number

Using the calibration routine as an example: Press the “Enter” key. The indicator display will show “Cal weight _” and the cursor will be blinking. The blinking cursor is the clue that you can enter an arbitrary number using the up, down, right, and enter keys. Pressing the up/down keys will scroll through the list (0 1 2 3 4 5 6 7 8 9 - .) in turn. When the desired number appears, press the right arrow (menu/help) key. The blinking cursor will advance one digit to the right, leaving your selected number in place. Continue this sequence until the desired numeric value is visible on the display. Press the “Enter” key to accept the value, or the “Cancel” key to abort.

Example: Enter a calibration weight of 25 pounds

- Start with the indicator at a normal weight display (“0.00 lb”)
- Press the “Menu/Help” key
- Scroll through the main menu using the up or down arrow keys until “Calibrate *” is displayed on the indicator
- Press the “Enter” key to start the calibration routine

- The indicator may display “Password” if a calibration password is required. If so, enter it (default calibration password is “Zero” “Zero” “Zero”)
- The indicator should now be displaying “Cal Weight” and a blinking cursor.
- Press the up arrow key. The display should now show “Cal weight 1”
- Press the up arrow key again. The display should now show “Cal Weight 2”
- Press the right arrow key to accept the first digit (2) and advance the blinking cursor to the next digit. The indicator should display “Cal weight 2_”
- Press the up arrow key five times to select a 5 as the second digit. The indicator should now display “Cal weight 25”
- Press the “Enter” key to accept 25 pounds as a calibration weight.
- The indicator will display “Cal-zero weight”. Press the “Cancel” key to abort the calibration process.

4.3 How to select from a list

This is very much like stepping through a menu. Some settings (such as displayed resolution) must be limited to one of several predetermined values. To edit one of these settings, press the “Enter” key. The currently selected value will move from the far right of the display to the left. This indicates that you may use the up and down arrow keys to scroll through a list of possible values for this setting. Once you’ve selected a value for the setting, press the “Enter” key to complete the selection process. As always, pressing the “Cancel” key will cancel the selection and restore the setting to the previous value.

5 General Scale Operations

5.1 Scale on procedure

Touch the “Zero / On” key. Indicator will come on and display will read “MicroWeigh by WeighTech” and then continue to the weigh mode. At this point the scale is ready for product or operator input.

5.2 Scale off procedure

To turn the scale off touch the “Menu / Help” key. The indicator will display “Power off *”. At this point touch the “Print / Enter” key and scale display will go blank, and the indicator will be off. (If the auto-on jumper is installed on the interface board, the indicator will immediately power up.)

5.3 Zero procedure

To zero the indicator touch the “Zero / On” key and the indicator will take a new zero. If the current weight reading is unstable, under capacity, or over capacity, no new pushbutton zero will be established.

5.4 Units procedure

To change the units of measure touch the “Units / Cancel” key. The units will change between pounds, kilograms, grams and ounces (assuming all the units are enabled in the “Parameter” menu) each time that the key is touched.

6 Machine Operation

6.1 Initial setup

- After powering up the indicators, you may see “Setup required” on the display. If so, go into the “Parameters” menu and set the scale capacity. “Cal required” means that the indicator has not been calibrated, and calibration must be done before the machine will operate. Once the capacity has been set and the indicator has been calibrated, set the indicator types.
- It is very important to ensure that the top two indicators are assigned to be remote heads and the bottom indicator assigned to be the master head. (This setting is accessed inside the “Dump settings” menu.) The top indicator should be set to “Remote 1”, the middle indicator to “Remote 2”, and the bottom (master) indicator should be set to “Master”. After changing the master/remote setting or the communications address, the indicator will need to be power cycled for the changes to take effect. (The indicator should automatically power cycle itself once you back all the way out of the menus.) If this system is a dual instead of a triple, set the “Mode” setting at the master indicator to “Dual”.
- At the master indicator, go to “Setup menu” / “I/O test”. Step through and toggle each output, verifying that the modules, pneumatic valves, cylinders, and indexing belt do the right thing. (The display will show you the current state—if the display shows that a gate is open, but it’s really closed, check to see if the plumbing is reversed.) Also check that the opto sensor is detecting boxes. (If the box detection is reversed, check the “Opto mode” setting in the “Dump settings” menu. The system can work with either light or dark operate sensors.)
- Select indexing method at the master indicator (“Dump settings” / “Indexer”) to match the machine: air clutch (or motor drive), stop gate, or coordinate with a Starflex bagger.
- The master indicator controls the dump settings and box indexing options. Adjust “Dump settings” / “Target” to the desired target weight of your box. The “Index time” setting (also in the “Dump settings” menu) controls how long the belt will

run after the edge of the box is detected. Adjust this value so that the box stops directly under the weigh hopper funnel. (There is a test function in the “Info menu” that will let you force a box to index.) “Index TO” should be set to 0. Tune the “Preindex” time to prevent the full box from moving away before the hopper is completely empty. If the box is still moving when the hopper begins to dump, add a little time to the “Postindex” setting. (The stop gate timing is slightly different. “Preindex” is the same, but the “Index time” is the time from stop gate down until the stop gate comes back up to catch the next empty box. Tune it so that the full box gets out of the way and the empty box is stopped by the gate. Adjust the “Postindex” so that the hopper doesn’t dump until the empty box is touching the stop gate.)

- The system can wiggle the diverter gates to clear minor jams. This requires adjusting four settings. Two settings (“Jam time” and “Jam weight”) are used to detect the jam. The logic behind the jam detection is that a hopper filling should be increasing in weight. If a period of time (“Jam time”) goes by without adding at least “Jam weight” to the hopper, the infeed must be jammed. Once a jam is detected, the system can wiggle the diverter gates to knock it free. The “Wiggle tm” setting controls how long the diverter gate will be open. Tune it so that the gate moves far enough to break up the jam, but not so long as to hold open and allow free flow into the other hoppers. The “Wiggle ct” setting is the number of times the diverters will wiggle once a jam is detected.
- Normally, the machine will wait for a stable weight before a hopper can dump. If for some reason the weights aren’t always stable, you can use the stable weight timeout (“Weigh TO”) setting to prevent the machine from hanging. This setting controls how long, in seconds, the machine will wait for the hopper to stabilize once target weight has been reached. If it doesn’t stabilize, the machine will accept the last unstable weight and let the hopper dump. 3 or 4 seconds would probably be a good starting place. (If the “Weigh TO” is set to zero, the hoppers won’t dump unless the weight stabilizes.)
- The system can also control the infeed, if the optional line stop output is connected. The “Line stop” timer controls how long the infeed is allowed to run (in seconds) when no hopper is filling. Adjust this time so that the infeed shuts off before product fills and overflows the top of the machine.
- There is a dump timeout setting. This can prevent the machine from hanging if a weigh hopper doesn’t fully empty out into the box. For example, if “Dump TO” is set to 5.00, the scale will wait up to 5 seconds for the weigh hopper to empty into the box and drop below zero threshold. If after 5 seconds the hopper hasn’t fully emptied out, it will give up and continue on. Set “Dump TO” to 0 to disable the timeout.
- Another new setting is the “Zero lmt”. This is the maximum weight that can be autozeroed out during the dump cycle. If the hopper doesn’t fully empty into the box, you may want to set the zero limit to prevent the scale from setting a new

zero with large pieces of product still in the hopper. Set “Zero lmt” to 0 to disable the limit and allow the scale to autozero any amount.

- If the master indicator is controlling an infeed belt, there is a “M1 mode” setting to select the output polarity. Use the I/O test to be sure the belt runs and stops when it should.

6.2 Starflex coordination

If the triple indexer is feeding a Starflex indexing belt and bagger, some settings will need to be changed. Set the “Indexer” setting to “Strflx” instead of “Clutch” and power cycle the master indicator. This will change the index output to a 250 ms pulse. The dump input signal is now expected to be a short pulse.

In operation the triple indexer will dump a hopper onto the belt, signal the Starflex to move the belt and fill a bag, then wait for a signal from the Starflex to dump the next hopper. If there is any concern that the Starflex might not be indexing on each pulse, you can break the hangup with the index timeout setting (“Index TO”). This setting controls how long the system will wait from the index signal until the next Starflex dump signal. If the timer expires with no signal from the Starflex, it would pulse the output again. You might end up with an empty bag, but that’s probably better than hanging up. Set “Index TO” to zero to disable the timeout and wait forever on the Starflex.

The I/O test menu will still work when in “Strflx” mode. One minor change affects the “Opto” display of the input module. Since the pulse from the Starflex is so short (1/10 of a second), the box / no box display is stretched to one second. You should see the display change every time the Starflex sends a signal. If not, verify wiring and polarity. (There are also counters in the info menu showing the number of pulses sent to the Starflex, number of pulses received, and the number of times the index timeout timer expired.)

6.3 Startup

Ensure that all hoppers are empty and that all moving parts are in good working condition. Power up the machine and indicators. No air cylinders should move when the indicators are powered on.

All three indicators can be placed into operation from the bottom, master, indicator. Touch the “Menu” key, scroll until “Production” is visible, and touch the “Enter” key. You can now scroll between the production options, so scroll until “Cycle on” is visible, and touch “Enter”. The display will show “Dump cycle on”, pause for a second or so, and return to the normal weight display. You should notice that the two remote indicators mounted above the master have also gone into the cycle on mode. The machine will now begin operation (buffer hopper gate will open and the weigh hopper gate will close). You may also notice that the left side of the display now shows “On”. This indicates that the machine is in the automatic weighing cycle.

6.4 Shutdown

From the normal weight display, touch the “Menu” key. Scroll up or down until you reach the “Production” menu. Touch the “Enter” key, and again scroll until “Cycle off” is visible on the display. Touch the “Enter” key. The display will show “Dump cycle off” for a few seconds, return to the normal weight display, and go into the off mode. In off mode, all hopper gates will close to prevent product flowthrough and the machine will not cycle. The left side of the display should show “Off” in this mode.

6.5 Cleanout

Select “Cycle idle” from the “Production” menu to enter idle mode. In idle mode, the hopper gates will all lock in the open position to allow cleaning. The indicator display will show “Idle” on the left side when the machine is in this mode. (If the machine is equipped with a washdown or idle button, you can push the switch in to place the machine in idle mode.)

6.6 Target weight

The target weight can be changed while the machine is in any operating mode, even while running production. Touch the “Menu” key, scroll until “Production” is shown on the display, touch “Enter”, and scroll until “Target” is visible. Once the target weight is visible on the display, touch the “Enter” key to modify it. Use the up or down arrows to adjust each digit in turn and press either “Enter” to set the new target weight, or “Cancel” to leave the target weight setting unchanged.

6.7 Dump cycle

The dump cycle is the operating cycle of the automatic weighing machine. The machine will attempt to fill the weigh hopper until target weight is exceeded. At this point, the buffer hopper gate will close. Once closed, the machine will pause to allow the weight to stabilize. As the product empties from the weigh hopper, the weight is monitored. Once the weight drops below the “Zero thr” setting, the weigh hopper gate will close. At this point in the operating cycle, the machine will pause for up to “Zero time” seconds to allow the weight to stabilize again (with the weigh hopper empty). If “Autozero” is turned on, the weight in the weigh hopper will be zeroed out. (Zeroing the weight at this point in the operating cycle helps keep dump weights accurate even if there’s some build up on the weigh hopper.) The cycle then repeats by opening the buffer hopper gate and filling the weigh hopper again.

6.8 Dry contact output option

There is an optional dry contact pulse output that will close for “Pulse time” seconds every “Pulse cnt” dumps. If either setting is zero the pulse output is disabled.

6.9 Claw option

If you're using the optional claw to hold the second box and a box stop to hold the box being filled, the "Claw delay" timer might need to be adjusted. It sets the delay between the box stop going down (releasing a full box) and the claw releasing the next box. It will control the gap between the full box and the next box. Tune it so that there's enough gap for the stop gate to reliably catch the next box each time.

6.10 Batching

The scale can batch (multiple dumps per box) if the target weight is set higher than the dump limit (hopper capacity). In this case, the box will index after multiple dumps.

6.11 Last box weight

The master indicator will display the last box weight for two seconds after the final dump of the box.

6.12 Downstream jam

The optional downstream jam detect opto is wired to position M6 on remote head 2. When the beam is solidly blocked for more than "Jam time" seconds, the remote will signal the master head that the downstream is jammed and no dumps will be allowed. As soon as the opto beam clears dumps will resume. Set "Jam time" to 0 to disable the downstream jam detection. There are debounce settings and an opto mode setting to allow for both light and dark operate sensors. (The downstream jam time, debounce, and opto mode settings are all in the "Parameters" menu at remote head 2.)

7 Calibration Procedure

7.1 Entering the calibration menu

With the indicator on and displaying weight, touch the "Menu / Help" key. The display will read "Power off *". Use the up / down arrows until the display reads "Calibrate *". Touch the "Print / Enter" key and the display should then show "Password". At this point key in the calibration password. (The default calibration password is "Zero" "Zero" "Zero" .)

7.2 Keying in cal weight

The display will show "Cal weight _" and the cursor will be blinking. Using the up, down, and right keys to enter the size of your calibration weight in pounds (i.e. 1, 2, 5, or 10). Press "Enter" to accept the cal weight, or "Cancel" if you make a mistake.

7.3 Calibration example

(Entering a 25.00 lb cal weight value.) The blinking cursor is the clue that you can enter an arbitrary number using the up and down keys. Pressing the up/down keys will scroll through the list (0 1 2 3 4 5 6 7 8 9 - .) in turn. When the desired number appears (2), press the right arrow “Menu / Help” key. The blinking cursor will advance one digit to the right (2 _), leaving your selected number in place. Continue this sequence until the desired numeric value is visible on the display (25_) (25._) (25.0_) (25.00). Press the “Enter” key to accept the value, or the “Cancel” key to abort.

7.4 Establishing a zero

The indicator will display “Cal–zero weight”. Clear the weighing platform of any foreign objects and once all vibration has died out, press the “Enter” key. Make sure that the platform is not disturbed during this process. Indicator will display “Zeroing...” as it takes an average reading of the zero offset weight (about three seconds).

7.5 Accepting a cal weight

The indicator will then display “Cal–add weight”. Add weight to the weighing platform (the weight should be the same amount as the keyed in cal weight) then touch the “Enter” key. The indicator will display “Scaling...” for about three seconds as it performs internal calculations. Finally, the indicator will display “Cal done” for about one second once the calibration cycle is complete.

8 Scale Parameters

To get to the parameters touch the “Menu/Help” key (indicator will display “Power off *”). Use the up or down arrows until the indicator displays “Setup Menu”. Touch the “Print/Enter” key and the indicator will prompt for a password. The password for this step will be as follows: starting from the left side of the keypad touch each key in turn from left to right. After entering the password the indicator will display “Parameters *”. At this point touch the “Print/Enter” key to access the parameters. Use the up and down arrows to scroll through and view each parameter.

8.1 “Units”

This parameter controls the setup unit of the indicator. Select from pounds (lb), kilograms (kg), grams (g), and ounces (oz). Once set, the indicator capacity, resolution, and calibration weights will be entered in this unit. The units parameter is both sealed and audited.

8.2 “Capacity”

Capacity sets the maximum capacity of the indicator, in setup units. This parameter is both sealed and audited. Factory default is 0, which must be changed before the

indicator will weigh.

8.3 “Resltn”

Parameter that sets the resolution of the indicator. Resolution is limited to values available on the scroll list. Resolution is set in terms of the setup units. This parameter is both sealed and audited.

8.4 “Stability”

This parameter controls how many consecutive weight readings are required to be within the motion sense band before the weight indication is considered to be stable. The indicator reads the analog input 7.5 Hz (7.5 times per second), so the default setting of four requires about a half second of stable weight. Either the net or gross light will come on when the weight is stable. This parameter is both sealed and audited.

8.5 “Motion sns”

Amount of motion, in divisions, allowed before the weight is considered unstable. Default is one division. This parameter is both sealed and audited.

8.6 “AZT”

Auto zero tracking on/off. This parameter is neither sealed nor audited. When on, stable weights within the “AZT band” of zero will automatically rezero the scale.

8.7 “AZT band”

Amount of weight, in divisions, that can be automatically zeroed out at one time. Default is 1 division. Parameter is sealed and audited.

8.8 “Calibrate”

This function starts the indicator calibration routine. It is sealed and audited. Refer to the calibration section of this manual for details.

8.9 “IZ set”

When this parameter is on, the indicator will attempt to establish a new initial zero every time the indicator powers on. HB44 limits the amount of weight that can be initially zeroed to 20% of scale capacity. (This initial zero does not reduce the indicator capacity.) This parameter is both sealed and audited.

8.10 “lb units”

Select on/off to enable or disable the pounds (lb) units when the Unit key is pressed in weighing mode. This parameter is both sealed and audited.

8.11 “kg units”

Select on/off to enable or disable the kilograms (kg) units when the Unit key is pressed in weighing mode. This parameter is both sealed and audited.

8.12 “g units”

Select on/off to enable or disable the grams (g) units when the Unit key is pressed in weighing mode. This parameter is both sealed and audited.

8.13 “oz units”

Select on/off to enable or disable the ounces (oz) units when the Unit key is pressed in weighing mode. This parameter is both sealed and audited.

8.14 “Defaults”

Restore all configuration parameters to factory default. This function is sealed and audited. Restoring factory defaults will require that the indicator be calibrated and reconfigured before it will weigh.

9 Display Messages

9.1 General warning/error messages

The following warning and error messages may appear at any time that the display is showing the current weight. They will not be visible when the indicator is displaying a menu item.

“Setup required” The indicator is still set to factory defaults, and will require configuration before entering service. This message will clear once the scale capacity (in “Setup Menu”/“Parameters”) has been set.

“Cal required” The indicator has not yet been calibrated. This message will clear once the scale has been calibrated.

“Excite Shorted!” The measured load cell excitation voltage has been below 1V for more than one second. To prevent any damage to the indicator or load cell, the excitation supply has been disabled. Double check the load cell cable and connections to the interface board, especially at the terminals marked “EX+” and “EX-”. The indicator must be turned off and back on again to clear this message.

“ADC Full Scale” The analog to digital converter reading the load cell went to full scale positive or negative for more than a second. This is usually caused by faulty wiring, or in severe cases, a seriously damaged load cell. Check connections, especially on the terminals marked “S1+” and “S1-”. This warning will clear automatically when the load cell readings come back into the normal range.

“Rmt 1 comm fail?” Remote indicator #1 has not replied to the master indicator. Check settings at remote #1 and verify that the RS-485 communications wiring is correct. The message will stop flashing once communications has been restored.

“Rmt 2 comm fail?” Remote indicator #2 has not replied to the master indicator. Check settings at remote #2 and verify that the RS-485 communications wiring is correct. The message will stop flashing once communications has been restored.

“Mstr comm fail?” (Remote indicators only). A remote indicator has not seen any communications from the master indicator. Check that the master indicator is powered up and configured as the master. Verify RS-485 communications wiring is correct. This message will stop flashing once communications has been restored.

“Downstream jam!” The second remote indicator opto sensor has detected a jam downstream of the scale. No dumps will be allowed until the jam has cleared. This message will clear once the opto beam is clear.

“IR locked.” Access to the setup menus is locked out. The “IR lock” setting in the parameter menu is turned on and it’s been more than one minute since the Palm was used to pull settings or totals. To unlock, use a Palm to pull totals or settings. You will then have 60 seconds to access the setup menus. If you do not wish to lock out the setup menus, turn off “IR lock”.

9.2 Calibration warnings

The following warning messages may appear when entering the calibration routine.

“Check load cell” This warning will only appear when entering the calibration routine. It indicates that the load cell reading is past full scale positive or negative. Calibration will not be allowed in this case. Refer to “ADC Full Scale” warning for troubleshooting information.

“Check excite” The load cell excitation supply was measured at less than 4.0 volts. Check the load cell and wiring, especially the terminals marked “EX+” and “EX-”. This warning can also be caused by excessive load cell current drain, such as from more than four load cells tied together, or a load cell with damaged strain gauges.

“Check ex- wire” Both load cell signal outputs were measured at more than 4.0 volts. This usually indicates that there’s a poor connection at the “EX-” terminal, or that the load cell excitation wiring has gone open (such as from physical damage to the cable).

“Check ex+ wire” Both load cell signal outputs were measured at less than 1.0 volts. This usually indicates that there’s a poor connection at the “EX+” terminal, or that the load cell excitation wiring has gone open (such as from physical damage to the cable).

“Check signals” The difference between the two load cell signal outputs was measured at more than 0.2 volts. This much imbalance between the two signals is usually caused by either incorrect wiring (such as swapping an excitation and signal wire), or by a severely bent load cell. Check connections, try swapping pairs, or replace the load cell.

“Check sig+ wire” The positive load cell signal was measured at either less than 1.5 volts or more than 4 volts. Check wiring at terminal marked “S1+”, check that the load cell signal and excite pairs are correct, and finally consider that the load cell may be damaged.

“Check sig- wire” The negative load cell signal was measured at either less than 1.5 volts or more than 4 volts. Check wiring at terminal marked “S1-”, check that the load cell signal and excite pairs are correct, and finally consider that the load cell may be damaged.

10 Menus

10.1 Main level

<i>Power off</i>	Turn off the indicator
<i>Production</i>	Enter the production submenu
<i>Washdown</i>	Disable keypad to prevent false keypresses during washdown
<i>Totals</i>	Display total weight, batch count, and average batch weight
<i>Calibrate</i>	Enter quick calibration routine
<i>Setup menu</i>	Bunch of stuff...see below
<i>Audit cfg</i>	Number of times an audited config parameter has been changed (HB44)
<i>Audit cal</i>	Number of times indicator has been calibrated (HB44)

10.2 Production

<i>Cycle on</i>	Start automatic dump cycle
<i>Cycle clean</i>	Close all hoppers, but continue indexing the Starflex bagger
<i>Cycle off</i>	Stop automatic dump cycle, close all gates
<i>Cycle idle</i>	Stop automatic dump cycle, leave all gates open for product flowthrough (failsafe)
<i>Target</i>	Target weight for complete batch, in pounds
<i>Totals</i>	Display total weight, batch count, and average batch weight
<i>Master</i>	Enable/disable the master hopper
<i>Rmt 1</i>	Enable/disable the remote 1 hopper
<i>Rmt 2</i>	Enable/disable the remote 2 hopper

10.3 Totals

<i>Cnt M</i>	Total number of boxes, master hopper
<i>Wgt M</i>	Total gross weight, master hopper
<i>Avg M</i>	Average box weight, master hopper
<i>Cnt 1</i>	Total number of boxes, remote 1 hopper
<i>Wgt 1</i>	Total gross weight, remote 1 hopper
<i>Avg 1</i>	Average box weight, remote 1 hopper
<i>Cnt 2</i>	Total number of boxes, remote 2 hopper
<i>Wgt 2</i>	Total gross weight, remote 2 hopper
<i>Avg 2</i>	Average box weight, remote 2 hopper
<i>Cnt *</i>	Total number of boxes, all hoppers
<i>Wgt *</i>	Total gross weight, all hoppers
<i>Avg *</i>	Average box weight, all hoppers
<i>Clear totals</i>	Clear accumulated totals (will ask you to confirm, touch “Enter” a second time to clear)

10.4 Setup menu

<i>Parameters</i>	Scale settings
<i>Dump settings</i>	Settings that control the automatic dump cycle operation
<i>I/O test</i>	Manually toggle outputs and read inputs
<i>RS-485 menu</i>	RS-485 communications counters
<i>Info menu</i>	Troubleshooting features
<i>Clock</i>	Set time/date
<i>Contrast</i>	Control display intensity

10.5 Parameters

<i>Setup units</i>	Setup units: used for entering capacity and resolution (defaults to pounds)
<i>Capacity</i>	Scale capacity, in setup units
<i>Resltn</i>	Scale resolution, in setup units
<i>Stability</i>	Number of consecutive readings required for stability
<i>Motion sns</i>	Number of divisions allowed before weight is considered unstable
<i>Filter</i>	Weighing filter speed: range of 0-0.9. Larger numbers make the filter slower, but weights are more stable
<i>AZT</i>	On/Off: Autozero tracking on/off, only affects weights near zero
<i>AZT band</i>	Amount of weight (in divisions) that can be zero tracked out
<i>Calibrate</i>	Start calibration routine
<i>IZ set</i>	Set initial zero at power up (default to off)
<i>lb units</i>	On/Off: Enables the units toggle key to include pound units
<i>kg units</i>	On/Off: Enables the units toggle key to include kilogram units
<i>g units</i>	On/Off: Enables the units toggle key to include gram units
<i>oz units</i>	On/Off: Enables the units toggle key to include ounce units
<i>Tare</i>	Enable/disable tare feature
<i>Address</i>	Communications/scale address (set remotes to 1 and 2)
<i>Cntst</i>	Enable/disable three key quick contrast adjustment
<i>IR lock</i>	When turned on, the Palm is required to access the setup menus
<i>Defaults</i>	Restore scale to factory default settings (all settings will be lost!)

10.6 Dump settings

<i>Target</i>	Target weight for complete batch, in pounds
<i>Dump lmt</i>	Target weight for each dump, in pounds (hopper capacity)
<i>Zero thrs</i>	Dump cycle zero threshold, in pounds (default is 5 pounds)
<i>Autozero</i>	Dump cycle autozero settings (on/off)
<i>Weigh TO</i>	How long to wait for a stable dump weight (0 waits forever)
<i>Zero time</i>	How long (in seconds) to wait for an autozero to happen
<i>Line stop</i>	How long (in seconds) to run infeed once all hoppers are full
<i>Empty time</i>	Extra time that weigh hopper will stay open after dropping below zero threshold
<i>Shock</i>	Number of consecutive readings above target weight before target is reached
<i>Filtered</i>	On/Off: Controls whether or not the target is compared to filtered or unfiltered weight
<i>Dump TO</i>	How long (in seconds) to wait for the hopper to empty into the box (0 disables TO)
<i>Zero lmt</i>	Max weight (in pounds) than can be autozeroed out during dump cycle (0 disables)
<i>Preindex</i>	Delay before the full box is indexed out
<i>Index</i>	Time from box detected until belt stops
<i>Index TO</i>	Max time (in seconds) to wait for a box edge to be detected. 0 disables.
<i>Postindex</i>	Time from box stop until hopper is allowed to dump
<i>Jam time</i>	Must see "Jam Weight" pounds in this many seconds during fill
<i>Jam weight</i>	Amount of weight that must be seen in "Jam Time" seconds during fill
<i>Wiggle tm</i>	Length of time the diverter will open and wiggle during a jam
<i>Wiggle ct</i>	Number of times diverter will wiggle when a jam is detected
<i>Dbnc +</i>	Number of ticks (100/second) required to detect leading edge of box
<i>Dbnc -</i>	Number of ticks (100/second) required to detect trailing edge of box
<i>Opto mode</i>	Opto sensor operating mode: select to match sensor type
<i>Indexer</i>	Select air clutch, Starflex coordination, or stop gate
<i>Mode</i>	Dual/triple mode selection
<i>Fill bypass</i>	Bypass fill when hopper is already full from wiggle (experimental)
<i>Fill pri</i>	Fill by priority rather than first ready (experimental)
<i>Dump pri</i>	Dump by priority rather than first ready (experimental)
<i>Antilock</i>	Try to detect and fix hopper hangs (experimental)
<i>Pulse time</i>	Length of pulse at dry contact output signal
<i>Pulse cnt</i>	Pulse dry contacts every X dumps
<i>Claw delay</i>	Delay from box stop down until claw release
<i>Claw dly 2</i>	(Pincher cycle only) if non-zero, claw fires after stop gate and this delay
<i>M1 mode</i>	Std: infeed belt runs when module is off
<i>Head</i>	Master/Remote: Set bottom head to master, other two heads to remote (must power2cycle for the change to take effect!)

10.7 I/O test

<i>Line</i>	Toggle the infeed line control
<i>Master dvrt</i>	Toggle the master diverter gate
<i>Master wgt</i>	Toggle the master weigh hopper gate
<i>Index belt</i>	Toggle the index belt
<i>Rmt #1 dvrt</i>	Toggle the diverter gate on remote #1
<i>Rmt #1 wgt</i>	Toggle the weigh hopper gate on remote #1
<i>Rmt #2 dvrt</i>	Toggle the diverter gate on remote #2
<i>Rmt #2 wgt</i>	Toggle the weigh hopper gate on remote #2
<i>Opto</i>	Live display of opto sensor state

10.8 RS-485 menu

<i>485 rx</i>	Number of bytes received
<i>485 tx</i>	Number of bytes transmitted
<i>485 in</i>	Number of commands received
<i>485 out</i>	Number of commands transmitted
<i>485 crc</i>	Number of corrupted commands received
<i>485 err</i>	Number of corrupt bytes received

10.9 Info menu

<i>Wiggle test</i>	Manually force the system to wiggle the diverters
<i>Fill jam</i>	Display the jam detection status
<i>Force index</i>	Manually force the system to index a single box without dumping (master only)
<i>Wgt M</i>	Live display of master weight
<i>Raw M</i>	Live display of master unfiltered weight
<i>Stb M</i>	Live display of master stability
<i>Wgt 1</i>	Live display of remote 1 weight
<i>Raw 1</i>	Live display of remote 1 unfiltered weight
<i>Stb 1</i>	Live display of remote 1 stability
<i>Wgt 2</i>	Live display of remote 2 weight
<i>Raw 2</i>	Live display of remote 2 unfiltered weight
<i>Stb 2</i>	Live display of remote 2 stability
<i>Stbl err</i>	Number of times a weigh hopper did not stabilize before the timeout aborted it
<i>Dump err</i>	Number of times a weigh hopper didn't empty out
<i>Dump</i>	Number of dump pulses received from the Starflex bagger
<i>Index</i>	Number of index pulses sent to the Starflex bagger
<i>Timeout</i>	Number of Starflex index timeouts
<i>ADC</i>	Raw counts display from analog to digital converter
<i>Offset</i>	Calibration zero offset, in raw counts
<i>App</i>	Name of firmware app (<i>tri_index_3</i>)
<i>Build</i>	Software revision info (<i>Build 49</i>)
<i>Date</i>	Date firmware was compiled (<i>02/13/2024</i>)
<i>Time</i>	Time of firmware compilation (<i>21:38:57</i>)
<i>Batt</i>	Current power supply/battery input voltage, in volts
<i>S1+</i>	Load cell #1 positive signal voltage (should be about half of excite voltage with good load cell)
<i>S1—</i>	Load cell #1 negative signal voltage (should be almost exactly the same as S1+ voltage)
<i>Excite</i>	Load cell excitation voltage (should be about 4.5V)
<i>Deadload</i>	Display platform deadload weight (assumes load cell has no offset)
<i>232 audit</i>	Transmit audit trail through RS-232 port (experimental)
<i>IZ autoreset</i>	Force a new initial zero
<i>IZ</i>	Current initial zero setting
<i>Debug msg</i>	On/Off: Turn this parameter on for more extensive messages during boot and dump cycle
<i>TX firmware</i>	Transmit firmware from master to remotes
<i>Bootload</i>	WeighTech use only

11 Detailed Menu Item Descriptions

11.1 Main menu level

Power off Turns off the MicroWeigh controller. When powered down, the indexing belts will continue to run to allow for easier cleanup.

Production Enter the production submenu. Most of the functions used on a daily basis are located in this menu, such as cycle on, off, and idle, the target weight, and the indexer totals menu.

Washdown Disable the keypad to prevent the touch sensitive keypad from registering false keypresses during a washdown period. To enable the keypad, play follow the leader and press the lit keys in order. You must get five in a row to exit washdown mode. (The controller will continue to operate normally in washdown mode—it only affects the keypad.)

Calibrate Quick calibration routine. The password required to enter the calibration routine is to touch the Zero/On key (far right side of keypad) three times. The indicator will first prompt for a calibration weight and the calibration weight units, then prompt for all weight to be removed, and finally for the calibration weight to be added. Calibration can only be performed when the system is in the cycle off mode. Before calibrating, several tests are run on the load cell and load cell connections. If any of these load cell tests fail, a warning message will displayed before the calibration weight prompt.

Setup menu Enter the setup submenu, where the configuration settings and test functions are located. This menu is password protected to prevent unqualified personnel from accessing the settings. The password is to touch each of the five keys in order, from left to right. (“Units”, “Menu”, “Print”, “Tare”, “Zero”).

Audit cfg Displays the number of times an audited configuration parameter has been changed. There is no way to reset this counter in the field.

Audit cal Displays the number of times the indicator has been calibrated. There is no method to reset this counter in the field.

11.2 Production menu

Cycle on Start the dump cycle. The master indicator will signal the two remote indicators to also go to the cycle on mode. All the diverter gates should shut, all the hopper gates should open, and then one diverter should open and begin filling a hopper. The hoppers will alternately fill and dump, with a box indexing for each dump.

Cycle off Exit the operating cycle. The master indicator will signal the two remote indicators to also go into the cycle off operating mode. This mode will shut all three diverter gates and the three weigh hopper gates.

Cycle idle Enter the idle/cleanout mode. The master indicator will signal the two remote indicators to also go into the cycle idle operating mode. This mode will open all gates and allow the indexing belt to run to aid in cleanup. This is also the default operating mode when the indicators are first powered on.

Target Target weight for the dump, in pounds.

Totals Enter the indexer totals menu. Totals can be viewed while the machine is in operation, and the display will update as each dump occurs. The totals are also available for download to a Palm through the front panel IR interface.

11.3 Totals menu

Tot wgt Total dump weight, in pounds.

Bat cnt Total number of batches/dumps.

Avg wgt Average dump weight.

Clear totals Clear all the totals. The indicator will prompt for confirmation (“Sure?”) before actually clearing the data. Press the Print/Enter key to clear, or the Units/Cancel key to abort.

11.4 Setup menu

Parameters Enter the parameter menu, where all the important (nonvolatile) settings are located. A word of warning—the “Default” option exists to return the settings to the factory defaults. It will prompt for confirmation, but once you’ve defaulted the settings any changes you’ve made are gone forever. If you don’t know why you’d need to return to factory defaults, you don’t.

Dump settings Enter the dump settings menu. Setting in this menu control the target and zero threshold weights, various timers, master/remote selection, and other machine specific variables.

Info menu Enter the info menu, which contains several interesting and useful nonadjustable parameters, such as the current firmware name and revision.

Clock Display current date and time. Also allows setting the real time clock. (The MicroWeigh controller real time clock is battery operated, so it will keep time even while the controller is powered off.)

Contrast Allows adjustment of the display intensity. Use the up and down arrows to scroll until the display contrast is nice and crisp, then press the Units/Cancel key to exit. This setting is retained through power cycles. There’s an emergency contrast adjustment that’s handy if the display is really washed out and hard to read: touch and hold the Units/Cancel and Menu/Help keys, then touch either the up or down key to adjust contrast. If you can’t tell whether to go up or down, touch the same key repeatedly until the display is easy to read. (It’ll loop around, so it doesn’t really matter if you use the up or down key.)

11.5 Parameters

Setup units Default units for resolution, capacity, and other internal settings. Default is to use pounds.

Capacity Maximum capacity of scale, in setup units. Must be set during initial installation to remove the “Setup Required” warning message from the display.

Resltn Displayed weight resolution, in pounds. Default is 0.05 pounds.

Stability Number of weight readings required to be within “Motion sns” divisions before the weight is considered stable.

Motion sns Size of band, in divisions, that weights must fall into to be considered stable. See “Stability” .

AZT (Automatic Zero Tracking) On/Off: when on, the scale will automatically zero out minor weight fluctuations need zero weight.

AZT band Number of divisions that AZT can automatically zero out.

Calibrate Perform a load cell test and calibration. (Feel free to enter the routine to do a load cell test, and touch the cancel key to exit without calibrating at the cal weight prompt. This can be a handy troubleshooting feature.)

IZ set Set initial zero. If you have to ask, you probably don’t need to be doing it.

lb units Enable or disable the pound units display. If enabled, the units key will allow the item weights to be displayed in pounds.

kg units Enable or disable the kilogram units display. If enabled, the units key will allow the item weights to be displayed in kilograms.

g units Enable or disable the gram units display. If enabled, the units key will allow the item weights to be displayed in grams.

oz units Enable or disable the ounce units display. If enabled, the units key will allow the item weights to be displayed in ounces.

Filter Filter control: Larger numbers will give a slower, more stable weight reading. Range is 0.01 to 0.99. This setting only affects the standard filter, not the super filter.

Address Communications address. Appears as part of the total and parameter information transmitted to the Palm through the IR port.

Defaults Set all settings back to factory defaults. Will prompt for confirmation before doing its business. “Sure?” touch the Units/Cancel key to abort, or the Print/Enter key to restore factory default settings. Defaulting will erase any changes to the parameters that have been made, so write them down before defaulting just in case!

11.6 Info menu

Force index Manually index one box. This is handy during setup and machine tuning to verify that the indexing belt is working properly and that the index time parameter is set to place the box under the hopper funnel.

ADC Live display of current raw unfiltered counts from analog to digital converter. Full scale is +/- 8 million counts, so values of more than several million are suspicious and could indicate a defective load cell or incorrect wiring.

Offset Current zero offset, in counts. Will update during an autozero cycle.

App Firmware application name (*tri_index_3*). Very handy to WeighTech personnel if you call and expect to receive accurate advice.

Build Firmware revision code (*Build 49*). Again, this is something that's good to know when contacting WeighTech with questions.

Date Date that firmware was compiled (*02/13/2024*). This is not the current date!

Time Timestamp of firmware compilation (*21:38:57*). This is not the current time!

Batt Current battery or power supply input voltage. Not really relevant with machine control installations, but will generally be between 6 and 8 volts.

S1+ Current load cell positive signal voltage. Should be about half of the excitation voltage, and very close to the negative load cell signal voltage.

S1- Current load cell negative signal voltage. Should be about half of the excitation voltage, and very close to the positive load cell signal voltage.

Excite Current excitation supply voltage, updated at about 1 Hz. Should be in the 4-5 volt range for normal operating. Very low readings could indicate cable damage, incorrect load cell wiring, or an excessive number of load cells connected to the indicator.

Deadload Display current calculated deadload weight, in pounds. Assumes that the load cell outputs exactly zero counts at no load, so consider this an approximation to the actual deadload.

232 audit Experimental RS-232 output of audit trail. WeighTech use only.

IZ autoset If enabled, the scale will attempt to set an initial zero every time it's powered on. Not really relevant to this firmware revision.

IZ Current initial zero setting.

Debug msg Not used in this firmware revision.

Bootload Enter bootloader for firmware update procedure. WeighTech use only.

12 Troubleshooting

12.1 Load cells

Go to the “Info menu” and verify that the “Excite” voltage is about 4.5V. A reading of less than 1V probably indicates a short from excite to ground. Confirm by removing the load cell connections. If the excite voltage reads normal with the load cell disconnected, you’ve got a short in the cable or a bad load cell.

Check to see that the signal voltage in the “Info menu” are about half of excite and equal. If one signal voltage is near zero, or near 4V, you may have a disconnected signal wire. Check that connection at the interface board. If the signal voltages are not near zero or 4V, but are more than a 0.5V different, you may have the load cell miswired, or a bent load cell.

If the indicator constantly shows “OVERLOAD” or “UNDERLOAD”, follow the instructions above. In addition, go to the “Info menu” and watch the “ADC” reading (raw counts). It shouldn’t vary more than 100-300 counts with a good load cell and a stable environment. With no load on the cell, it should be within +/- 10,000 counts of zero. (Deadload can cause the no load reading to shift.) If the no load reading is really large (say, greater than one million counts or less than negative one million counts) and the connections are solid, you probably have a bent load cell.

Unstable or noisy weights? Perform all the steps listed above. A really good test is to temporarily disconnect the load cell and substitute a known good load cell simulator (available for purchase from WeighTech), or a known good load cell. Calibrate the scale with a convenient test weight and check to see if the weight reading is stable. If so, the noisy load cell has probably been damaged or water-soaked. If the indicator still displays a noisy weight with a load cell simulator, the problem may be in the indicator. Contact WeighTech for further assistance.

12.2 Banner opto sensors

The Banner opto sensors will act strangely if wired incorrectly. Be certain that you connect the brown wire (12V) last when rewiring sensors with power on. If the brown and blue wires are swapped, or if the blue wire (ground) is connected last, the sensor will go into alarm mode. In alarm mode, the sensor will appear to work but the MicroWeigh controller will not receive the correct signal. If the led indicators on the Banner appear to be working correctly, but the MicroWeigh front panel indicators don’t show any changes when the beam is blocked, check the Banner sensor wiring.

12.3 Machines

Most machine related problems can be traced to bad air supply (excessive water and condensation), sticky air cylinders, worn or clogged solenoid air valves, and wiring. A quick test is to put the machine in “Cycle off” mode—all the valves should energize, shutting all the gates. Then put the machine in “Cycle idle” mode, which will de-energize all the valves and open all the gates. If a gate doesn’t move during these two tests, start swapping parts around to determine where the problem lies.

To examine the dump cycle details, turn on the “Debug flag” in the “Info menu”. The main weight display will show the current state of the dump cycle on the far left side of the display. Refer to the descriptions of each state below to determine if there’s a problem with gates or communications.

12.4 Dump cycle state machine

holding Diverter and hopper gate are both closed. Hopper is waiting for the fill command from the master indicator, which will allow it to open the diverter gate and continue to the filling state.

filling Diverter gate open, hopper gate closed. Indicator is monitoring the weight as the hopper fills, waiting for target weight to be reached. Once target weight has been reached, the dump cycle will advance to the weighing state.

weighing Both gates are closed. The indicator is waiting for the hopper weight to stabilize. Once the weight is stable, the dump cycle will go into the box state. (If the weight does not stabilize in “Weigh TO” seconds, the last known weight will be used instead.)

box Both gates are closed. Hopper is ready to dump. Indicator is waiting for the master indicator to coordinate with the indexer. When the dump signal arrives, the cycle will step to the dumping state.

dumping Diverter gate closed, weigh hopper gate open. The indicator is comparing the weight in the hopper to the zero threshold—once it drops below zero threshold, the dump cycle will advance to the emptying state. If the “Dump TO” value is greater than 0, the system will not wait more than dump timeout seconds for the hopper to empty out.

emptying Diverter gate closed, hopper gate open. Indicator remains in the state for “Empty Time” seconds to ensure that all the product has time to fall. Once the empty time timer expires, the dump cycle will be in the zeroing state.

zeroing Both gates are closed. The indicator is attempting to autozero (if enabled in the “Dump Settings” menu). If a stable weight can’t be reached before the zero time timer expires, the zero error counter will increment. The indicator will return to the holding state as soon as a stable autozero weight is reached or the zero timer expires. (If “Zero lmt” is greater than 0, only autozero if the hopper weight is less than zero limit.)

idle Both gates are open. The indicator is in the cleanout/cycle idle operating mode. The dump cycle will switch to the holding state when “Cycle On” is selected, or into the off state if “Cycle Off” is selected.

off Both gates are closed. The indicator is in the cycle off operating mode. The dump cycle will switch to the holding state when “Cycle On” is selected, or into the idle state if “Cycle Idle” is selected.

12.5 Before calling WeighTech...

Write down a few key pieces of information. Gather the indicator serial number from the front panel, the software application name and build number from the “Info menu”, and grab the current settings if you have access to a Palm. If anything on the indicator has changed, been replaced, or been modified, mention that to the service technician too. If the problem involves fill rates, hangs, or questions regarding machine capabilities, be ready to describe the product, product flow rate, and any bag/box/tote/combo sizes. If you’re calling about unstable weight readings, over/underload, or other load cell related problems, have the ADC, excite, and signal readings from the “Info menu” handy. When calling, be prepared to describe what is wrong (“it doesn’t work!” isn’t a good description–“hopper gate doesn’t shut in off mode” is much better) and what you expected to see.

12.6 Module assignments (master indicator)

Slot	Module Type	Function	Module on (lit)	Module off (dark)
M1	ODC5 (red)	Infeed drive	Belt stop	Belt run
M2	OAC5 (black)	Diverter gate	Divert	Fill
M3	OAC5 (black)	Weigh gate	Shut	Open
M4	IDC5 (white)	Box sensor	No box	Box
M5	ODC5 (red)	Index drive	Belt stop	Belt run
M6				

12.7 Module assignments (remote indicators)

Slot	Module Type	Function	Module on (lit)	Module off (dark)
M1	OAC5 (black)	Diverter gate	Divert	Fill
M2	OAC5 (black)	Weigh gate	Shut	Open
M3				
M4				
M5				
M6				

Parameter Listing

E3 FS 1 460V 1.0HP V3.11

Infeed

ID	Value	Description	Visible
P-01	60.0 Hz	Maximum Frequency / Speed Limit	✓
P-02	0.0 Hz	Minimum Frequency / Speed Limit	✓
P-03	1.0 s	Acceleration Ramp Time	✓
P-04	0.5 s	Deceleration Ramp Time	✓
P-05	0: Ramp to Stop (Mains Loss Ride-Through)	Stop Mode Select	✓
P-06	0: Disable	Energy Optimiser	✓
P-07	480 V	Motor Rated Voltage	✓
P-08	1.7 A	Motor Rated Current	✓
P-09	60 Hz	Motor Rated Frequency	✓
P-10	0 rpm	Motor Rated Speed	✓
P-11	3.0 %	Torque Boost	✓
P-12	0: Terminal Mode	Primary Command Source	✓
P-13	0: Industrial Mode	Industrial / Fan-Pump Control Mode	✓
P-14	201	Extended Menu Access Code	✓

ID	Value	Description	Visible
P-15	10: [NO/Close to Run] [NC/Open to Stop] [Analog Input 1/Preset 1] [Analog Input 1]	Digital Inputs Function Select	✓
P-16	0: 0-10V	Analog Input 1 Format	✓
P-17	1: 8 kHz	Effective Switching Frequency	✓
P-18	1: Drive Healthy	User Relay Output Function Select	✓
P-19	100.0 %	Adjustable Relay threshold	✓
P-20	0.0 Hz	Preset / Jog Frequency / Speed 1	✓
P-21	25.0 Hz	Preset / Jog Frequency / Speed 2	✓
P-22	40.0 Hz	Preset / Jog Frequency / Speed 3	✓
P-23	60.0 Hz	Preset / Jog Frequency / Speed 4	✓
P-24	0.00 s	2nd Ramp Time (Fast Stop)	✓
P-25	8: Motor Speed	Analog Output Function Select	✓
P-26	0.0 Hz	Skip Frequency Hysteresis Band	✓
P-27	0.0 Hz	Skip Frequency Centre Point	✓
P-28	0 V	V/F Characteristic Adjustment Voltage	✓
P-29	0 Hz	V/F Characteristic Adjustment Frequency	✓
P-30	0: EdgE-r	Terminal Mode Restart Function	✓
P-30	0: N.C. (Fire Mode Active on Open)	Fire Mode Input Logic	✓
P-30	0: Input Not Latched	Fire Mode Input Latch	✓
P-31	1: Previous Keypad Speed	Keypad / Modbus Mode Restart Function	✓
P-32	0.0 s	DC Injection Duration	✓
P-32	0: DC Injection on STOP	DC Injection Braking Mode	✓
P-33	0: Disabled	Spin Start Enable	✓
P-34	0: Disabled	Brake Chopper Enable	✓
P-35	100.0 %	Analog Input 1 Scaling / Slave Speed Scaling	✓

P-36	1	Drive Address	✓
P-36	6: 115.2kpbs (Modbus)/500kpbs (CANopen)	Modbus / CANbus baudrate	✓
P-36	4: t 3000ms	Communications Trip Delay	✓
P-37	101	Extended Menu Access Code Definition	✓
P-38	0: Unlocked	Parameter Access Lock	✓
P-39	0.0 %	Analog Input 1 Offset	✓
P-40	0.000	Display Speed Scaling Factor	✓
P-40	0: Motor Speed	Display Scaling Source	✓
P-41	1.0	User PI Proportional Gain	✓
P-42	1.0 s	User PI Integral Time Constant	✓
P-43	0: Direct Operation (Wake up from 0%)	User PI Operating Mode	✓
P-44	0: Digital	User PI Reference Select	✓
P-45	0.0 %	User PI Digital Reference (Setpoint)	✓
P-46	0: 2nd Analog Input (T4)	User PI Feedback Select	✓
P-47	0: 0-10V	Analog Input 2 Format	✓
P-48	0.1 s	'Standby' Activation Delay Time	✓
P-49	5.0 %	PI Wake Up Error Level	✓
P-50	0.0 %	User Relay Output Hysteresis	✓
P-51	1: V/F Speed Control	Motor Control Mode	✓
P-52	0: Disable	Motor Parameter Auto-tune	✓
P-53	50.0 %	Vector Mode Speed Controller Gain	✓
P-54	150 %	Maximum Current Limit	✓
P-55	8.00 Ohm	Motor Stator Resistance	✓
P-56	650.0 mH	Motor Stator d-axis Inductance (Lsd)	✓
P-57	65.0 mH	Motor Stator q-axis Inductance (Lsq)	✓
P-58	0.0 Hz	DC Injection Speed	✓
P-59	20.0 %	DC Injection Current	✓
P-60	1: Enable	Thermal Overload Retention	✓
P-60	0: Disable	Thermal Overload Management	✓
P-61	0: Disabled (Services are disabled unless P-62 = 0)	Ethernet Service Option	✓
P-62	0 Mins	Ethernet Service Timeout	✓
P-63	0: Standard	Modbus Mode Selection	✓
P-64	0: Terminal 2 Or Switch Forward Or Switch Reverse	IP66 DI1 Source	✓
P-65	0: Terminal 3 Or Switch Reverse	IP66 DI2 Source	✓
P-66	0.0 %	Analog Output Limit	✓

ID	Value	Description
P0-01	100.0 %	1st Analog input value
P0-02	0.0 %	2nd Analog Input Value
P0-03	60.0 Hz	Speed Reference Value
P0-04	DI-1: 0 DI-2: 0 DI-3: 0 DI-4: 1	Digital Inputs Status
P0-05	0.0 %	User PI Output
P0-06	1 V	DC Bus Voltage Ripple
P0-07	0 V	Motor Voltage
P0-08	621 V	DC Bus Voltage
P0-09	22 C	Heatsink Temperature
P0-10	0:00:01	Run Time Since Date of Manufacture
P0-11	0:00:01	Run Time since Last Trip 1
P0-12	0:00:01	Run Time since Last Trip 2
P0-13	0:00:00	Load factory default Trip Log

	parameters (P-dEF)	
	0:00:00	No Fault (no-Flt)
	0:00:00	No Fault (no-Flt)
	0:00:00	No Fault (no-Flt)
P0-14	0:00:01	Run Time Since Last Enable
P0-15	0 V	DC Bus Voltage Log
	0 V	
	0 V	
	0 V	
	0 V	
	0 V	
	0 V	
	0 V	
P0-16	0 C	Heatsink Temperature Log
	0 C	
	0 C	
	0 C	
	0 C	
	0 C	
	0 C	
	0 C	
P0-17	0.0 A	Motor Current Log
	0.0 A	
	0.0 A	
	0.0 A	
	0.0 A	
	0.0 A	
	0.0 A	
	0.0 A	
P0-18	0 V	DC Bus Ripple Log
	0 V	
	0 V	
	0 V	
	0 V	
	0 V	
	0 V	
	0 V	
P0-19	0 C	Internal Drive Temperature Log
	0 C	
	0 C	
	0 C	
	0 C	
	0 C	
	0 C	
	0 C	
P0-20	35 C	Internal Drive Temperature
P0-21	0000	CANopen Process Data Input
	0000	
	0000	
	0000	
P0-22	0000	CANopen Process Data Output
	0000	
	0000	
	0016	
P0-23	0:00:00	Accumulated Time with Heatsink > 85C
P0-24	0:00:00	Accumulated Time with Drive Internal Temperature > 80C

P0-25	0.0 Hz	Estimated Rotor Speed
P0-26	0.0 kWh 0 MWh	kWh Meter / MWh Meter
P0-27	0:00:18	Total Run Time of Drive Fans
P0-28	I/O V3.11 72B9 Power V3.11 0E61	Software Version and Checksum
P0-29	Drive type: E3 Drive rating: 460V 3~ 1.0HP Firmware version: 3.11 Scope capture data available: No	Drive Type Identifier
P0-30	649453 / 18 / 057	Drive Serial Number
P0-31	Id: 0.0 Iq: 0.0	Motor Current Id / Iq
P0-32	1: 8 kHz	Actual PWM Switching Frequency
P0-33	0	Critical Fault Counter – O-I
P0-34	0	Critical Fault Counter – O-Volts
P0-35	1	Critical Fault Counter – U-Volts
P0-36	0	Critical Fault Counter – O-temp (Heatsink)
P0-37	0	Critical Fault Counter – b O-I (chopper)
P0-38	0	Critical Fault Counter – O-hEAt (control)
P0-39	4	Modbus Comms Error Counter
P0-40	255	CANbus Comms Error Counter
P0-41	0	I/O Processor Comms Errors
P0-42	0	Power Stage uC Comms Errors
P0-43	0:11:22	Drive Power Up Time (Life Time)
P0-44	Offset 0 Ref 0	Phase U Current Offset & Ref
P0-45	Offset 0 Ref 0	Phase V Current Offset & Ref
P0-46	Offset 0 Ref 0	Phase W Current Offset & Ref
P0-47	0:00:00 0	Fire Mode Total Activation Time and Fire Mode Activation Counter
P0-48	1 0 2 0	Scope Channel 1 & 2
P0-49	3 0 4 0	Scope Channel 3 & 4
P0-50	L 3.07 b 2.00	Bootloader and Motor Control

Parameter Listing

E3 FS 1 460V 2.0HP V3.11

BOX TAKEAWAY

ID	Value	Description	Visible
P-01	12.0 Hz	Maximum Frequency / Speed Limit	✓
P-02	0.0 Hz	Minimum Frequency / Speed Limit	✓
P-03	0.5 s	Acceleration Ramp Time	✓
P-04	0.5 s	Deceleration Ramp Time	✓
P-05	0: Ramp to Stop (Mains Loss Ride-Through)	Stop Mode Select	✓
P-06	0: Disable	Energy Optimiser	✓
P-07	230 V	Motor Rated Voltage	✓
P-08	3.4 A	Motor Rated Current	✓
P-09	60 Hz	Motor Rated Frequency	✓
P-10	0 rpm	Motor Rated Speed	✓
P-11	15.0 %	Torque Boost	✓
P-12	0: Terminal Mode	Primary Command Source	✓
P-13	0: Industrial Mode	Industrial / Fan-Pump Control Mode	✓
P-14	201	Extended Menu Access Code	✓

ID	Value	Description	Visible
P-15	10: [NO/Close to Run] [NC/Open to Stop] [Analog Input 1/Preset 1] [Analog Input 1]	Digital Inputs Function Select	✓
P-16	0: 0-10V	Analog Input 1 Format	✓
P-17	1: 8 kHz	Effective Switching Frequency	✓
P-18	1: Drive Healthy	User Relay Output Function Select	✓
P-19	100.0 %	Adjustable Relay threshold	✓
P-20	0.0 Hz	Preset / Jog Frequency / Speed 1	✓
P-21	12.0 Hz	Preset / Jog Frequency / Speed 2	✓
P-22	12.0 Hz	Preset / Jog Frequency / Speed 3	✓
P-23	12.0 Hz	Preset / Jog Frequency / Speed 4	✓
P-24	0.00 s	2nd Ramp Time (Fast Stop)	✓
P-25	8: Motor Speed	Analog Output Function Select	✓
P-26	0.0 Hz	Skip Frequency Hysteresis Band	✓
P-27	0.0 Hz	Skip Frequency Centre Point	✓
P-28	0 V	V/F Characteristic Adjustment Voltage	✓
P-29	0 Hz	V/F Characteristic Adjustment Frequency	✓
P-30	0: EdgE-r	Terminal Mode Restart Function	✓
P-30	0: N.C. (Fire Mode Active on Open)	Fire Mode Input Logic	✓
P-30	0: Input Not Latched	Fire Mode Input Latch	✓
P-31	1: Previous Keypad Speed	Keypad / Modbus Mode Restart Function	✓
P-32	0.0 s	DC Injection Duration	✓
P-32	0: DC Injection on STOP	DC Injection Braking Mode	✓
P-33	0: Disabled	Spin Start Enable	✓
P-34	0: Disabled	Brake Chopper Enable	✓
P-35	100.0 %	Analog Input 1 Scaling / Slave Speed Scaling	✓

P-36	2	Drive Address	✓
P-36	6: 115.2kbps (Modbus)/500kbps (CANopen)	Modbus / CANbus baudrate	✓
P-36	4: t 3000ms	Communications Trip Delay	✓
P-37	101	Extended Menu Access Code Definition	✓
P-38	0: Unlocked	Parameter Access Lock	✓
P-39	0.0 %	Analog Input 1 Offset	✓
P-40	0.000	Display Speed Scaling Factor	✓
P-40	0: Motor Speed	Display Scaling Source	✓
P-41	1.0	User PI Proportional Gain	✓
P-42	1.0 s	User PI Integral Time Constant	✓
P-43	0: Direct Operation (Wake up from 0%)	User PI Operating Mode	✓
P-44	0: Digital	User PI Reference Select	✓
P-45	0.0 %	User PI Digital Reference (Setpoint)	✓
P-46	0: 2nd Analog Input (T4)	User PI Feedback Select	✓
P-47	0: 0-10V	Analog Input 2 Format	✓
P-48	0.0 s	'Standby' Activation Delay Time	✓
P-49	5.0 %	PI Wake Up Error Level	✓
P-50	0.0 %	User Relay Output Hysteresis	✓
P-51	2: PM Vector Speed Control	Motor Control Mode	✓
P-52	0: Disable	Motor Parameter Auto-tune	✓
P-53	50.0 %	Vector Mode Speed Controller Gain	✓
P-54	150 %	Maximum Current Limit	✓
P-55	4.20 Ohm	Motor Stator Resistance	✓
P-56	350.0 mH	Motor Stator d-axis Inductance (Lsd)	✓
P-57	35.0 mH	Motor Stator q-axis Inductance (Lsq)	✓
P-58	0.0 Hz	DC Injection Speed	✓
P-59	20.0 %	DC Injection Current	✓
P-60	1: Enable	Thermal Overload Retention	✓
P-60	0: Disable	Thermal Overload Management	✓
P-61	0: Disabled (Services are disabled unless P-62 = 0)	Ethernet Service Option	✓
P-62	0 Mins	Ethernet Service Timeout	✓
P-63	0: Standard	Modbus Mode Selection	✓
P-64	0: Terminal 2 Or Switch Forward Or Switch Reverse	IP66 DI1 Source	✓
P-65	0: Terminal 3 Or Switch Reverse	IP66 DI2 Source	✓
P-66	0.0 %	Analog Output Limit	✓

ID	Value	Description
P0-01	100.0 %	1st Analog input value
P0-02	0.0 %	2nd Analog Input Value
P0-03	12.0 Hz	Speed Reference Value
P0-04	DI-1: 0 DI-2: 0 DI-3: 0 DI-4: 1	Digital Inputs Status
P0-05	0.0 %	User PI Output
P0-06	1 V	DC Bus Voltage Ripple
P0-07	0 V	Motor Voltage
P0-08	633 V	DC Bus Voltage
P0-09	23 C	Heatsink Temperature
P0-10	0:01:15	Run Time Since Date of Manufacture
P0-11	0:01:13	Run Time since Last Trip 1
P0-12	0:01:13	Run Time since Last Trip 2
P0-13	0:00:00	Load factory default Trip Log

	parameters (P-dEF)	
	0:00:00	No Fault (no-Flt)
	0:00:00	No Fault (no-Flt)
	0:00:00	No Fault (no-Flt)
P0-14	0:00:01	Run Time Since Last Enable
P0-15	0 V	DC Bus Voltage Log
	0 V	
	0 V	
	0 V	
	0 V	
	0 V	
	0 V	
	0 V	
P0-16	0 C	Heatsink Temperature Log
	0 C	
	0 C	
	0 C	
	0 C	
	0 C	
	0 C	
	0 C	
P0-17	0.0 A	Motor Current Log
	0.0 A	
	0.0 A	
	0.0 A	
	0.0 A	
	0.0 A	
	0.0 A	
	0.0 A	
P0-18	0 V	DC Bus Ripple Log
	0 V	
	0 V	
	0 V	
	0 V	
	0 V	
	0 V	
	0 V	
P0-19	0 C	Internal Drive Temperature Log
	0 C	
	0 C	
	0 C	
	0 C	
	0 C	
	0 C	
	0 C	
P0-20	36 C	Internal Drive Temperature
P0-21	0000	CANopen Process Data Input
	0000	
	0000	
	0000	
P0-22	0000	CANopen Process Data Output
	0000	
	0000	
	0017	
P0-23	0:00:00	Accumulated Time with Heatsink > 85C
P0-24	0:00:00	Accumulated Time with Drive Internal Temperature > 80C

P0-25	0.0 Hz	Estimated Rotor Speed
P0-26	0.0 kWh 0 MWh	kWh Meter / MWh Meter
P0-27	0:00:22	Total Run Time of Drive Fans
P0-28	I/O V3.11 72B9 Power V3.11 0E61	Software Version and Checksum
P0-29	Drive type: E3 Drive rating: 460V 3~ 2.0HP Firmware version: 3.11 Scope capture data available: No	Drive Type Identifier
P0-30	654310 / 24 / 083	Drive Serial Number
P0-31	Id: 0.0 Iq: 0.0	Motor Current Id / Iq
P0-32	1: 8 kHz	Actual PWM Switching Frequency
P0-33	0	Critical Fault Counter – O-I
P0-34	0	Critical Fault Counter – O-Volts
P0-35	1	Critical Fault Counter – U-Volts
P0-36	0	Critical Fault Counter – O-temp (Heatsink)
P0-37	0	Critical Fault Counter – b O-I (chopper)
P0-38	0	Critical Fault Counter – O-hEAt (control)
P0-39	4	Modbus Comms Error Counter
P0-40	255	CANbus Comms Error Counter
P0-41	0	I/O Processor Comms Errors
P0-42	0	Power Stage uC Comms Errors
P0-43	0:15:35	Drive Power Up Time (Life Time)
P0-44	Offset 4008 Ref 4006	Phase U Current Offset & Ref
P0-45	Offset 4018 Ref 0	Phase V Current Offset & Ref
P0-46	Offset 3986 Ref 0	Phase W Current Offset & Ref
P0-47	0:00:00 0	Fire Mode Total Activation Time and Fire Mode Activation Counter
P0-48	1 0 2 0	Scope Channel 1 & 2
P0-49	3 0 4 0	Scope Channel 3 & 4
P0-50	L 3.07 b 2.00	Bootloader and Motor Control

15 Replacement Parts

Part Number	Description
1000-10	Load cell, 200 lb capacity
10575	Motor, PM (index, buffer)
11377	Sprocket, stainless steel, 12T (index, buffer)
AA0005	Flow control, GR-QS-6 (inline)
AA0006	Air tubing, duo blue/black, 6 mm x 50 m
AA0009	Air regulator
AA0026	Rod clevis, M10 x 1.25 mm, stainless steel
AC0001	Air cylinder, weigh hoppers
AC0002	Air cylinder, diverter (25-150)
AF0001	Air fitting, QS-1/8-6 (pack of 10)
AF0006	Air fitting, QSL-1/8-6 (pack of 10)
AF0018	Air fitting, QSL-6 (pack of 10)
AV0044	Air valve and coil assembly (Festo)
AV0045	DIN plug, air valve assembly (Festo)
AV0046	LED gasket, air valve assembly (Festo)
AV0047	Substation, air valve assembly (Festo)
AV0048	End plate kit, air valve assembly (Festo)
CB0060	Belting, series 800, flattop, 16 " wide (buffer)
CB0117	Sprocket, 800 Series, 7.7 PD, 12T, 1.5" square bore (index, buffer)
CB0126	Belting, series 800, 16" wide with special flight arrangement (24" center-4" center-24" center)
EF0009	Strain relief, MicroWeigh
EP0028	Variable speed drive, 460 VAC, 1 HP (ODE3)
EP0029	Variable speed drive, 460 VAC, 2 HP (ODE3)
HW0018	MicroWeigh housing screw (pack of 4)
HW0019	Screw, 6-19 x 0.375, trilobe PPH, MicroWeigh (pack of 10)
SE0038	Cord set, right angle connector (MQDC-430RA)
SE0065	Photo electric receiver, detachable (T18-2VPRL-Q8)
SE0066	Photo electric emitter, detachable (T18-2NAEL-Q8)
SR0001	Input relay (white, G4-IDC5)
SR0002	Output relay (black, G4-OAC5)
SR0003	Output relay (red, G4-ODC5)
SS0006	Stop switch assembly w/ contacts
SS0020	Pushbutton, momentary, green (10250T30G)
SS0024	Pushbutton, momentary, red (10250T30R)
WE0028	Main gasket, MicroWeigh
WE0029-1	Power cord, detachable, MicroWeigh
WE0046-101	Front housing assembly, MicroWeigh machine control (specify firmware name and revision)
WE0047-101	MicroWeigh machine control back housing assembly w/ board
WE0047-102	MicroWeigh machine control back housing (with board)
WE0076-101	Machine control interface board assembly, MicroWeigh



JOB: TRIPLE INDEXER

PLANT: BUDDY'S KITCHEN

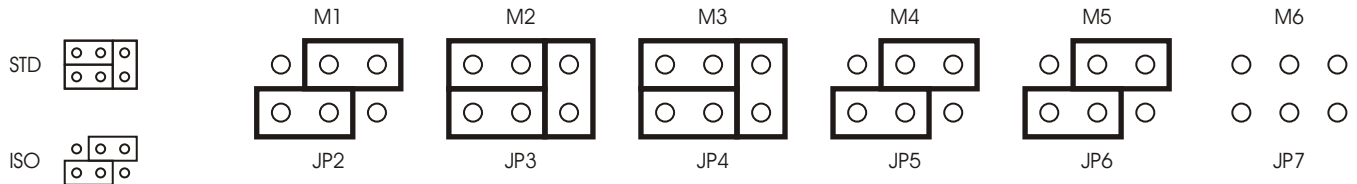
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PAGE: 1 OF 5

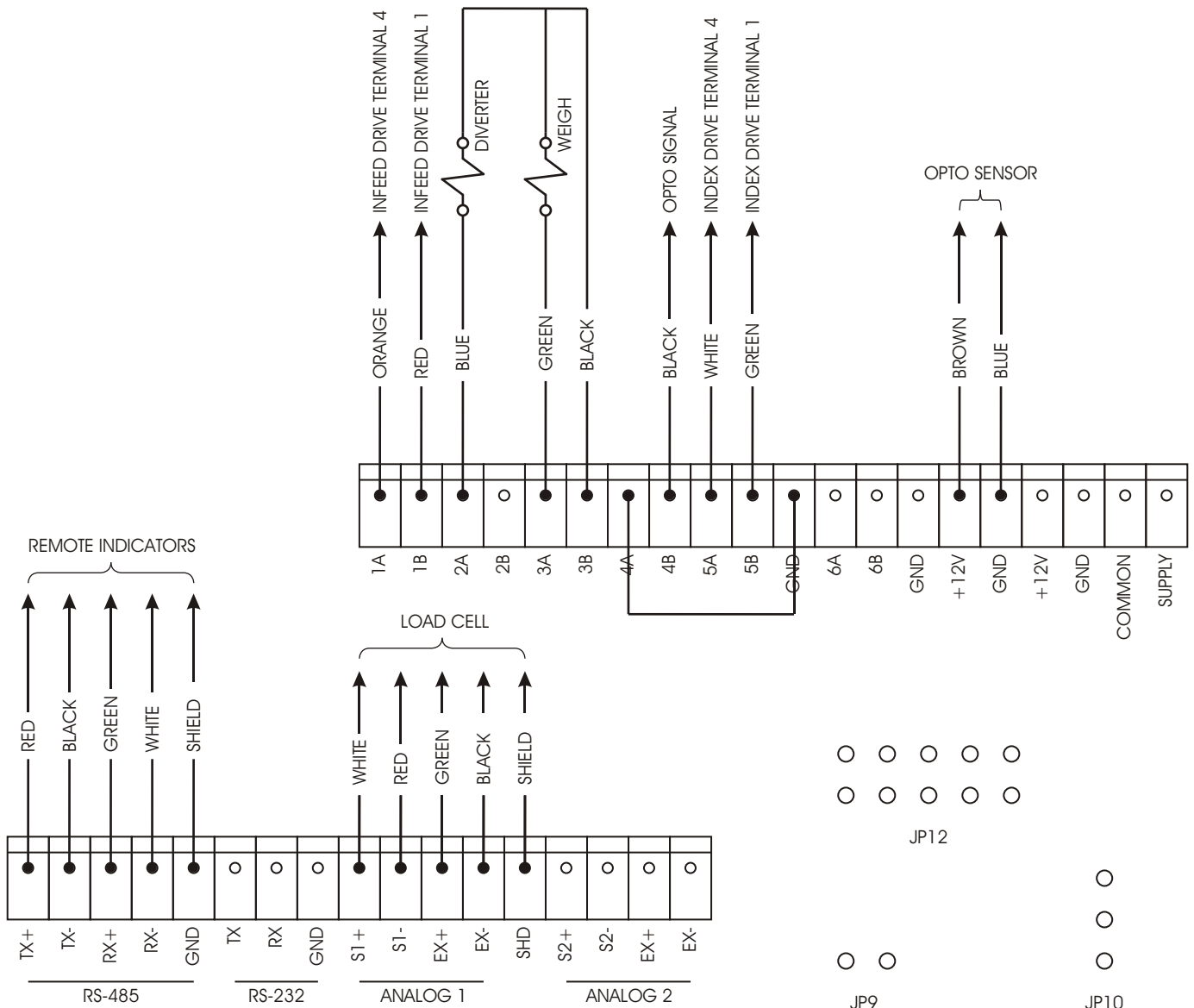
DRAWN BY: NEWELL

DATE: 4-26-2026

FIRMWARE: TRI_INDEX_3 (MASTER INDICATOR)



SLOT	TYPE	FUNCTION	LIT	DARK
M1	RED ODC-5	INFEED DRIVE	STOP	RUN
M2	BLACK OAC-5	DIVERTER	SHUT	OPEN
M3	BLACK OAC-5	WEIGH GATE	SHUT	OPEN
M4	WHITE IDC-5	OPTO SENSOR	NO BOX	BOX
M5	RED ODC-5	INDEX BELT	STOP	RUN
M6				





JOB: TRIPLE INDEXER

PLANT: BUDDY'S KITCHEN

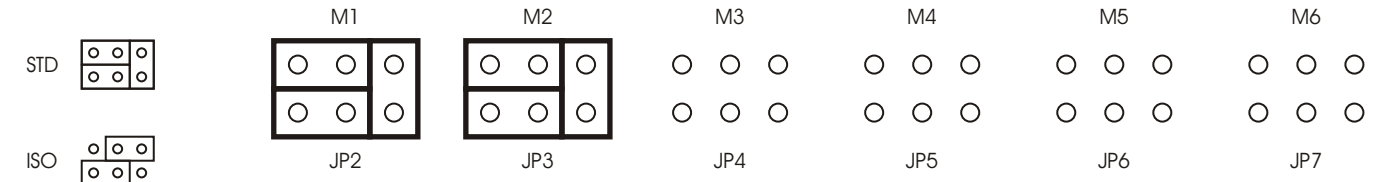
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PAGE: 2 OF 5

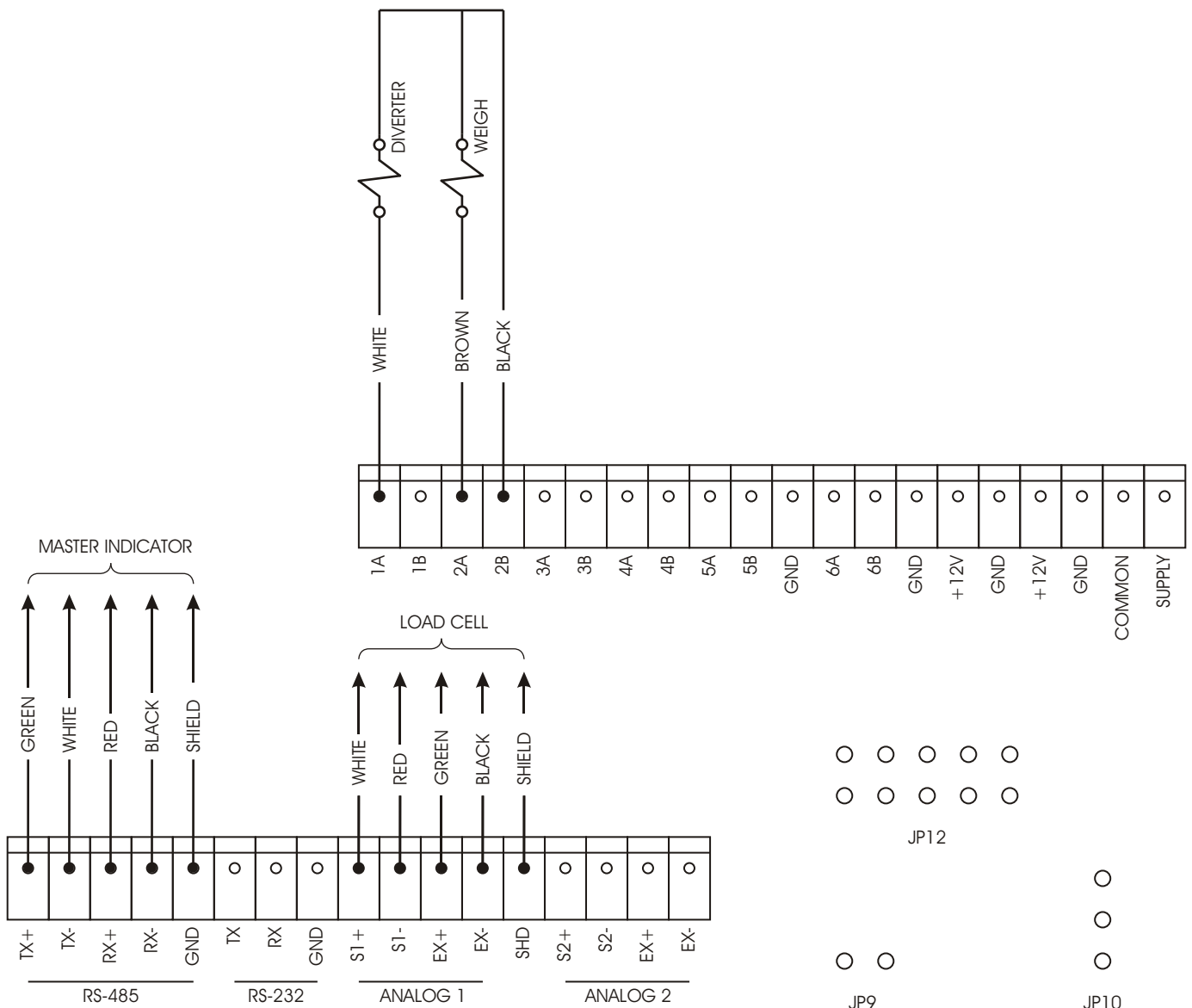
DRAWN BY: NEWELL

DATE: 4-26-2026

FIRMWARE: TRI_INDEX_3 (REMOTE INDICATOR 1)



SLOT	TYPE	FUNCTION	LIT	DARK
M1	BLACK OAC-5	DIVERTER	SHUT	OPEN
M2	BLACK OAC-5	WEIGH GATE	SHUT	OPEN
M3				
M4				
M5				
M6				





JOB: TRIPLE INDEXER

PLANT: BUDDY'S KITCHEN

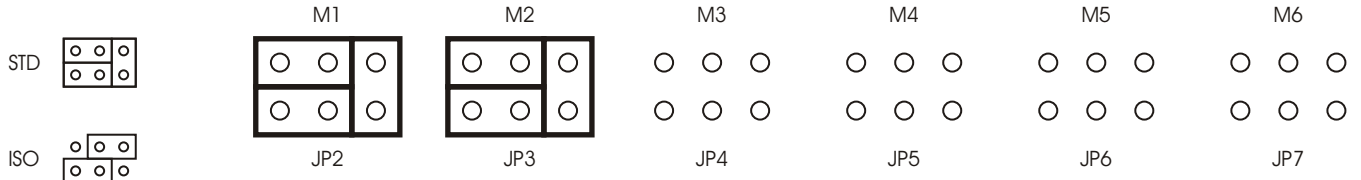
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PAGE: 3 OF 5

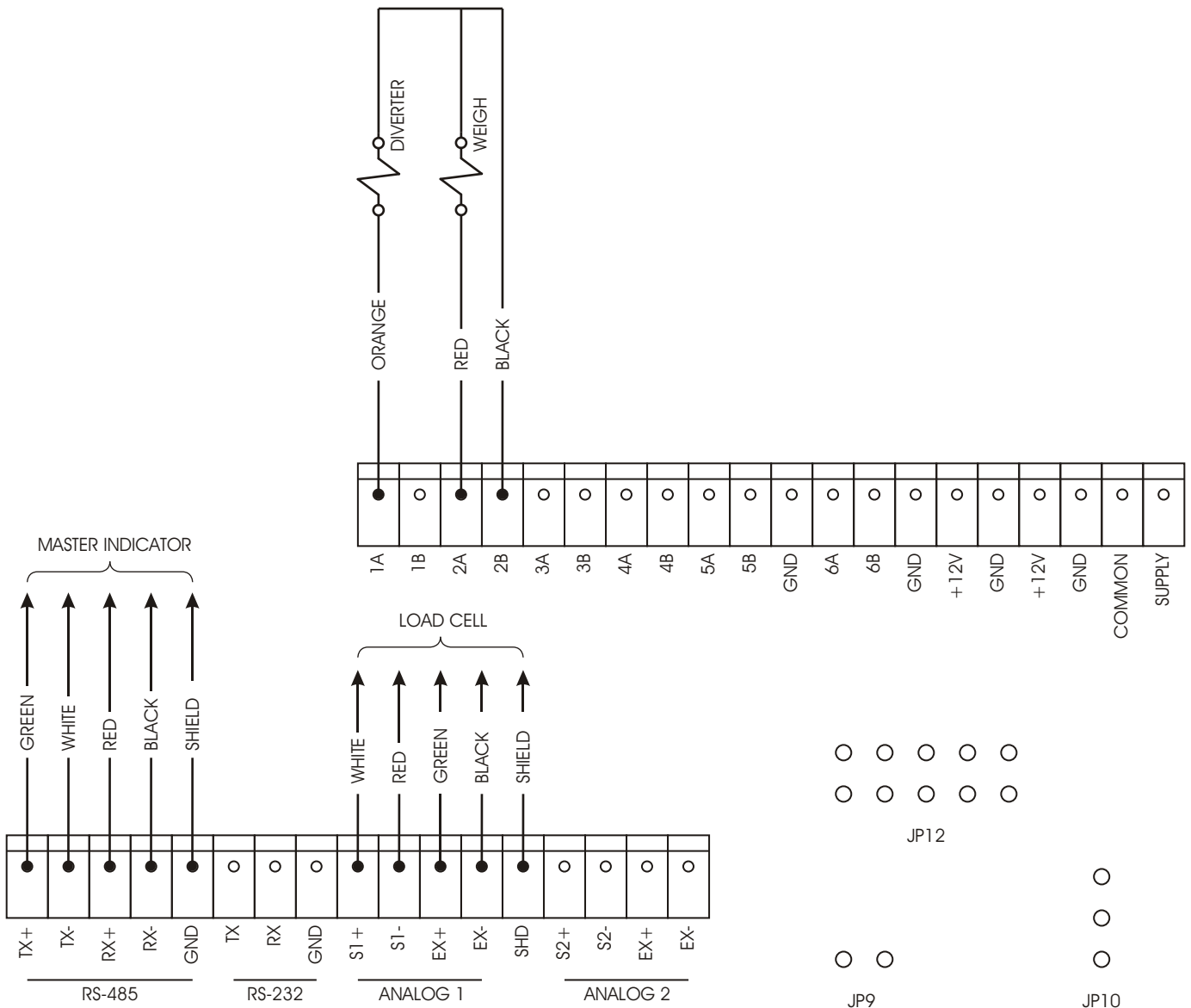
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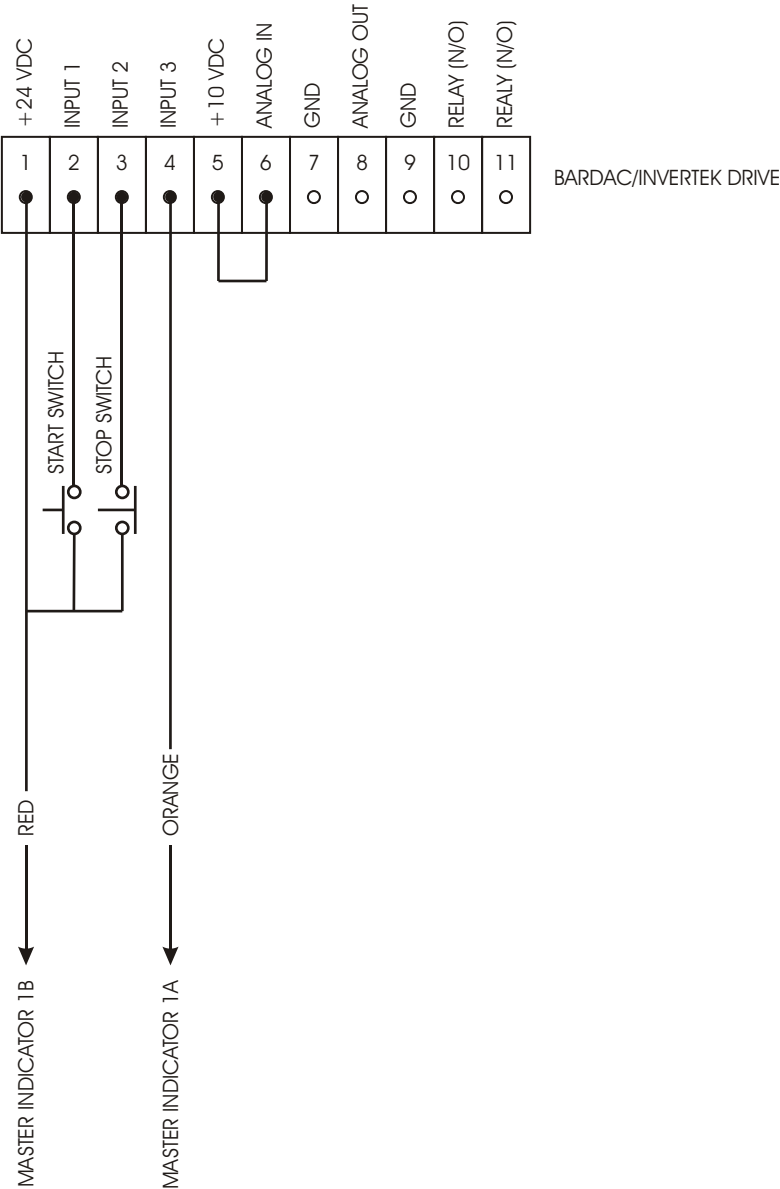
DATE: 4-26-2026

FIRMWARE: TRI_INDEX_3 (REMOTE INDICATOR 2)



SLOT	TYPE	FUNCTION	LIT	DARK
M1	BLACK OAC-5	DIVERTER	SHUT	OPEN
M2	BLACK OAC-5	WEIGH GATE	SHUT	OPEN
M3				
M4				
M5				
M6				

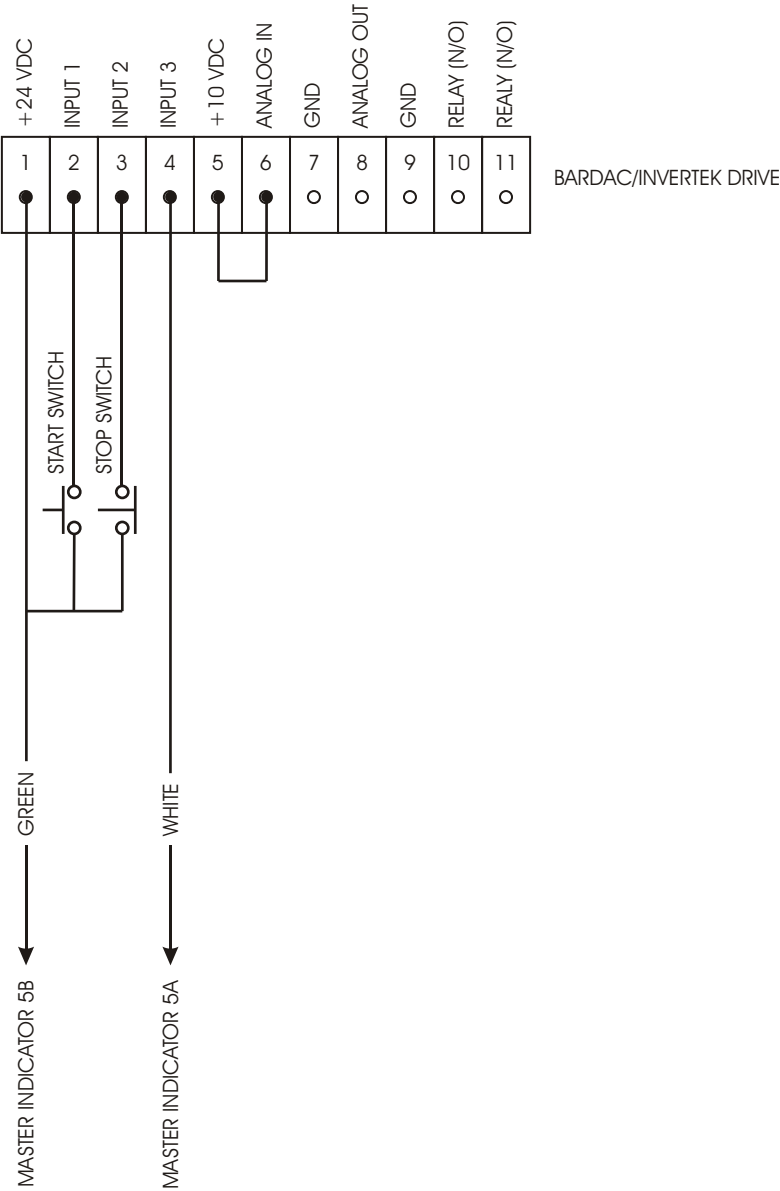






JOB: TRIPLE INDEXER
PLANT: BUDDY'S KITCHEN
LOCATION: LAKEVILLE, MN
PAGE: 5 OF 5
DRAWN BY: NEWELL
NOTES: INDEX DRIVE WIRING

DATE: 4-26-2026



17 Load Cell Color Codes

Manufacturer	Models	Signal +	Signal -	Excite +	Excite -	Shield	Sense +	Sense -
Advanced Transducers		Green	White	Red	Black	Bare wire		
Allegany Technology		Red	White	Green	Black	Bare wire		
Artech		Green	White	Red	Black			
Beowulf		White	Red	Green	Black			
BLH	C2P1 C3P1 T2P1 T3P1	White	Red	Green	Black	Yellow		
Cardinal		White	Red	Green	Black	Bare wire		
Celtron	CSB DSR LOC SQB STC STC-SS DSR CLB HED DLB LPS HOC MOC	Green Green Red	White White White	Red Red Green	Black Blue Black	Bare wire Bare wire Bare wire		
Dillon	Canister Tension Compression Z-cell	Black Black White	Red Red Green	Green White Red	White Green Black	Orange Orange Orange		
Force Measurement		Green	White	Red	Black	Bare wire		
GSE		White	Green	Red	Black	Bare wire		
HBM	BLC BLF JRT PWS RSC SBF SB3 USB U1T Z6 BBS PLC B35 SP4	White White Green White	Red Red White Red	Green Green Red Green	Black Black Black Black	Yellow Bare Yellow Yellow	Orange	Blue
Interface	SSM 1200 3200	Green	White	Red	Black	Bare wire		
Kubota		Green	Blue	Red	White	Yellow		
National		White	Red	Green	Black	Yellow		
NCI		White	Green	Red	Black	Bare wire		
Pennsylvania		Green	White	Orange	Blue	Bare wire		
Phillips		Green	Grey	Red	Blue	Bare wire		
Revere Transducer	62HU 63HU 363 953 9523 92CC 93CC 42U 43U 263D 462 5102 5103 5123 5223 5723 6762 9102 9103 9123 9363 392B 642 652 692B2 BSP HPS USP1 792 933 SHB SSB CP1 CSP1 RLC	Green Green White White White Brown	White White Red Red Red White	Red Red Green Green Green Pink	Black Black Black Black Black Grey	Bare wire Orange Bare wire Clear Black Bare wire		
Rice Lake	RL20000 RL20000SS RL20001 RL20001HE RL30000 RL35023 RL35023S RL35082 RL35082S RL35083 RL39123 RL39523 RL50210 RL65044 RL70000 RL75016 RL75016SS RL75040A RL75058 RL75060 RL75223 RL90000 RLETB RLETS RLHSS RLMK4 RL50500 RL70000SS RL71000HE RL75016HE RLMK15 RLMK21 RL75061 RLMK1 RL1521	Green White Green White	White Green White	Red Red Red	Black Black Blue	Bare wire Orange Bare wire	Yellow	Blue
Sensotec	White	Green	Red	Black	Bare wire			
Sensortronics	60001 60008 60018 60030 60036 60040 60048 60048SS 60050 60051 60060 60060-0101 60063 65007 65016 65016SS 65016W 65023 65023S 65023SS 65024 65040A 65040S 65058 65058S 65061A 65083 65083S 65114 60007 60064 65088-1000 65088-1114	Green White White	White Red Red	Red Green Green	Black Black Black	Bare wire Bare wire Orange		
Tedea Huntleigh	4158 3411 3421 240 1010 1022 1040 1042 1140 1250 1260 1320 9010 605 1030 1240 1241 355 620 3510	Green Green Red	White White White	Red Red Green	Black Black Black	Bare wire Bare wire Bare wire	Blue Blue	Brown Brown
Toledo		White	Red	Blue	Black	Bare wire	Green	Grey
Weigh-Tronics		White	Red	Green	Black	White/Orange		

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